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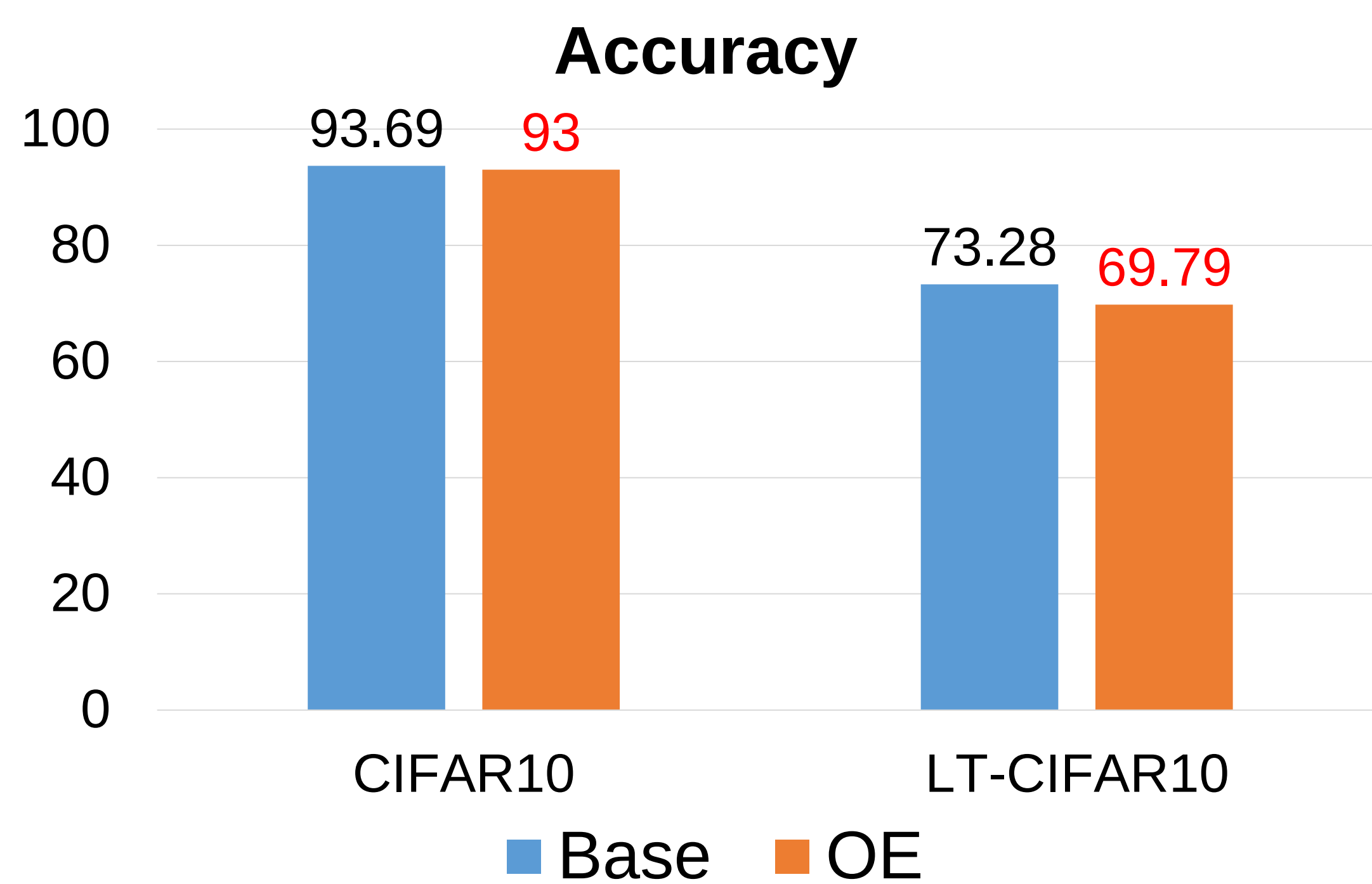
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Introduction

- In out-of-distribution (OOD) detection, fine-tuning with auxiliary outlier data often improves detection performance at the cost of classification accuracy. (**OOD-accuracy trade-off [1]**)
- To establish a more practical and effective paradigm, we optimize three critical factors: model reminder, data sampling, and representation learning.
- We propose: (1) Self-Knowledge Distillation (SKD) to mitigate accuracy reduction; (2) Semi-hard Outlier Sampling (SOS) to improve detection efficiency with minimal data; and (3) Outlier-aware Supervised Contrastive Learning (OSCL) to promote ID-OOD separability. Optimizing these factors produces cumulative gains, boosting both OOD detection performance and classification accuracy.
- Our framework outperforms existing methods across diverse benchmarks, particularly in long-tailed scenarios, providing a robust baseline for real-world OOD detection.

Motivation

- Why **Outlier Exposure (OE [1])** regularization drops classification accuracy

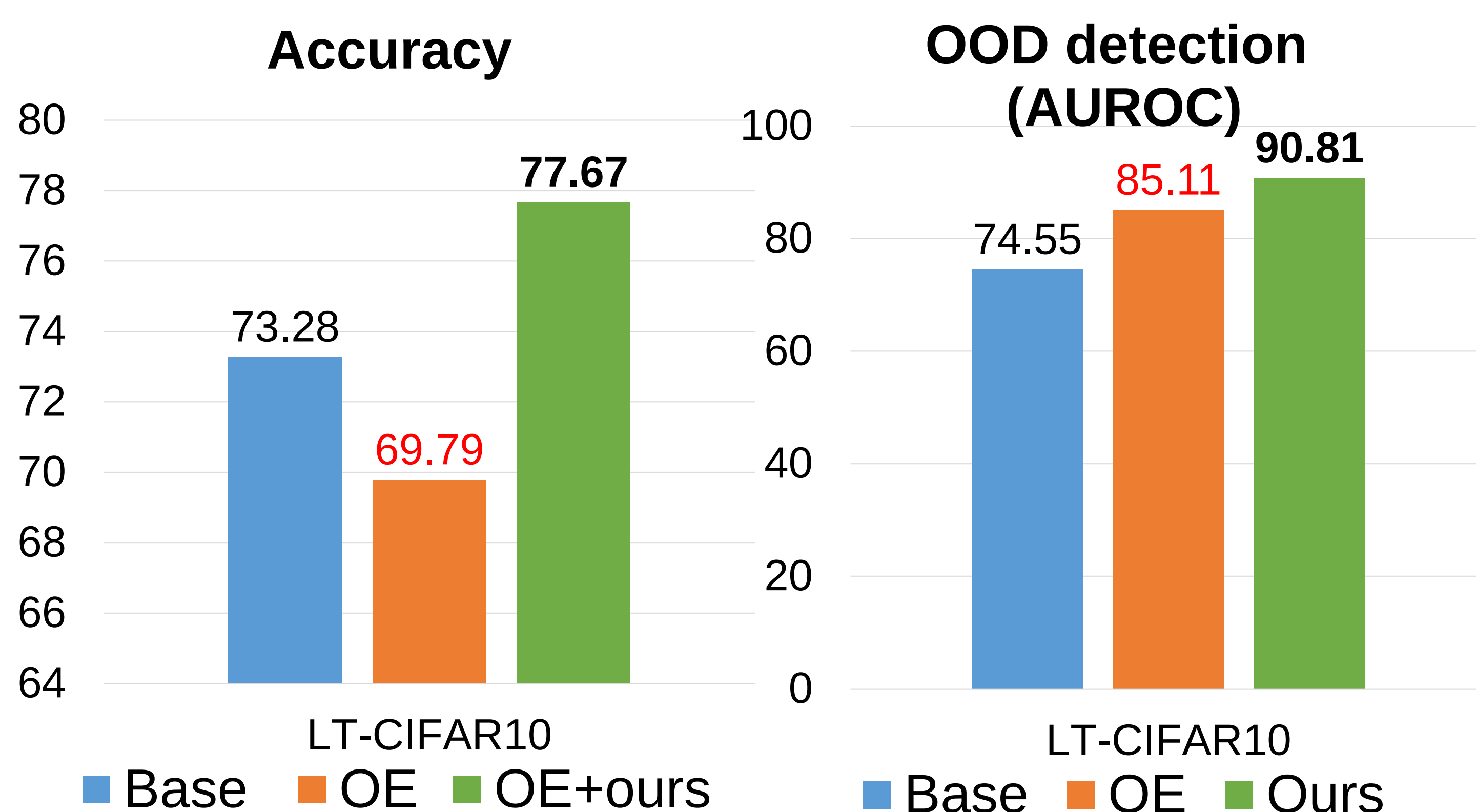


- Main reason: **Loss of original distribution for in-distribution**

Optimizing Three Critical Factors

- **Model** : Reminder of original model
- **Data** : outlier training data quality
- **Loss** : representation learning quality

Key Result



References

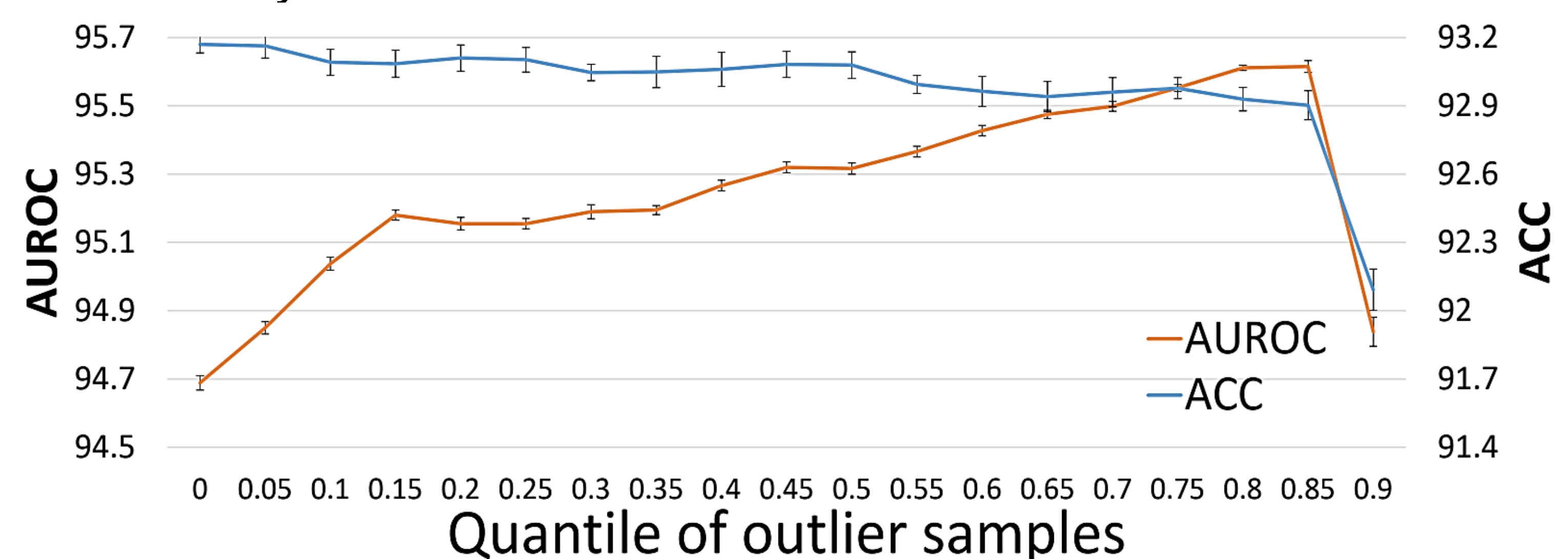
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(1) Self-Knowledge Distillation (SKD)

- Problem: Standard OE fine-tuning suppresses the original ID distribution, causing classification accuracy degradation.
- Self-Knowledge Distillation (SKD): Uses the original pre-trained model as a frozen teacher to preserve ID distribution during fine-tuning.
- Model Reminder: The student is optimized with a **soft-target distillation loss [5]**, referencing fixed predictions from the teacher.

(2) Semi-hard Outlier Sampling (SOS)

- Observation: Only 10% (30K) of TinyImages-300K is sufficient to achieve strong OOD detection performance.
- Hardness Analysis: Increasing outlier hardness improves OOD detection, but degrades ID classification accuracy, especially for the hardest outliers.
- Semi-hard Outlier Sampling (SOS): Selects outliers from an optimal hardness quantile range to balance OOD detection and ID accuracy.

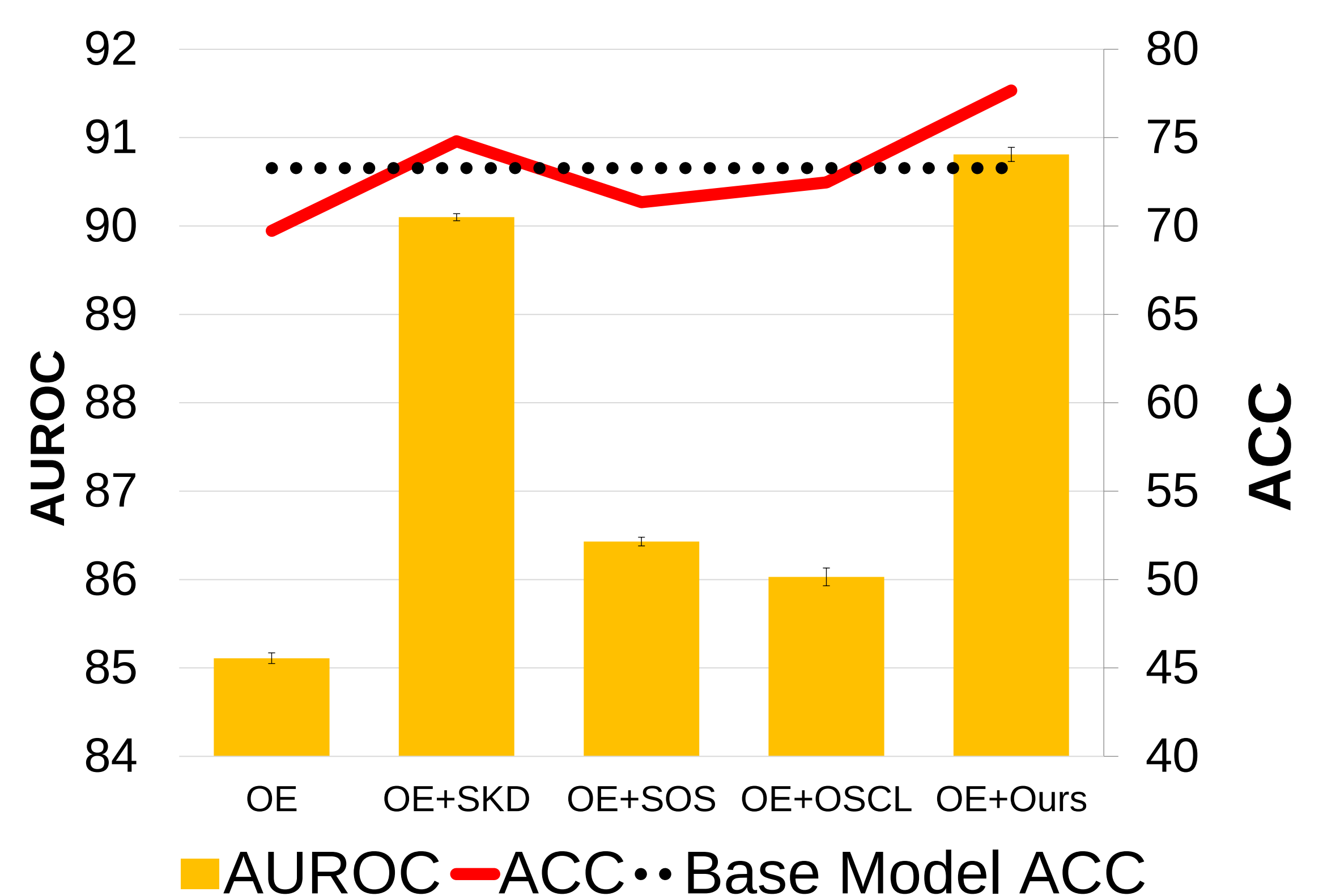


(3) Outlier-aware Supervised Contrastive Learning

- Limitation: Naively applying **existing contrastive learning [6]** provides only limited OOD improvements.
- Outlier-aware SCL (OSCL): Uses outlier samples as explicit negatives, pushing OOD representations away from ID clusters.
- Multi-batch Transform: Generates multiple augmented views per sample to further improve representation quality.

Experimental Result

- OOD detection and Accuracy in LT-CIFAR10 (**OE [1]**)



- Ours with other finetuning loss (**OECC[2],Energy[3],B-Energy[4]**)

